



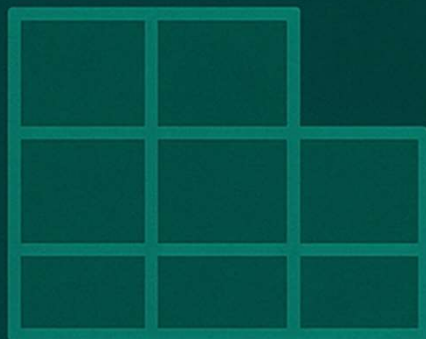
Introduction to Check the Model Performance

Supervised Machine Learning: Classification

Accuracy

Log Loss

CONFUSION MATRIX



Specificity



Precision



Recall



F1 Score

F1 Score



Performance Measurement of Classification Models

Introduction

When building a classification model (like predicting if a transaction is fraudulent or if a patient has cancer), it's important to evaluate how well the model is performing. Just getting "accurate" predictions is not always enough. Let's walk through the key performance metrics, step by step, with simple examples so anyone can understand them.

Accuracy Score

Definition: Accuracy is the percentage of correct predictions made by the model out of all predictions.

Formula:

$$\text{Accuracy} = \frac{\text{Total Number of Correctly Classified Points}}{\text{Total Number of Data Points}}$$

Example:

If a model correctly predicts 90 out of 100 test results, the accuracy is 90%.

Problem With Accuracy Score

- **Imbalanced Data:** If 95 out of 100 transactions are non-fraud and only 5 are fraud, a model that predicts everything as non-fraud will still have 95% accuracy. But it's **completely missing the fraud cases**.
- **Ignoring Prediction Confidence (Probability):** Suppose two models give the following predictions:

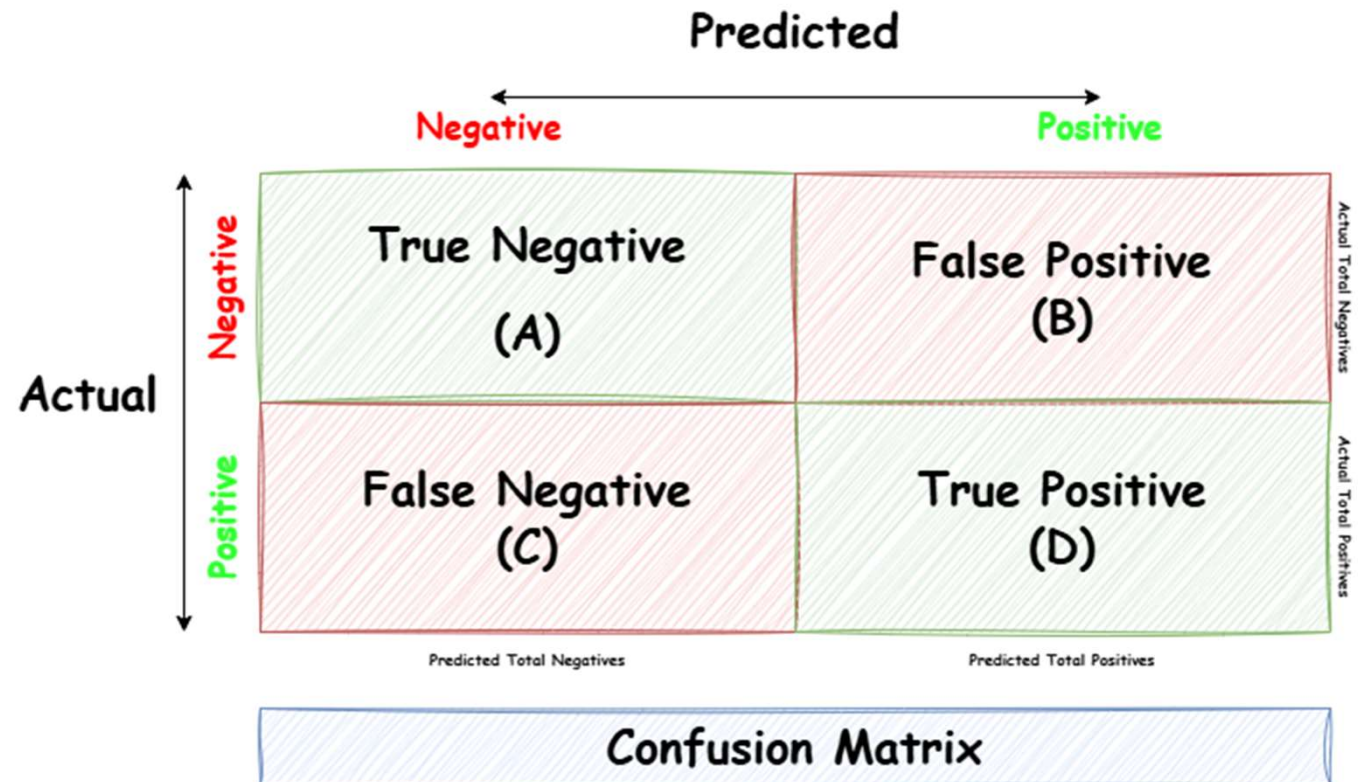
Example	Actual	Model A Probability	Model B Probability	Model A Prediction	Model B Prediction
1	1	0.90	0.60	1	1
2	1	0.80	0.55	1	1
3	0	0.10	0.45	0	0
4	0	0.15	0.48	0	0

- Both models give the same predictions (1, 1, 0, 0) — hence same accuracy — but Model A is more confident and thus better.
- Conclusion: Accuracy alone cannot compare models or reflect performance on imbalanced data.

Confusion Matrix

Definition: A table that is often used to describe the performance of a classification model on a set of data for which the true values are known. It summarizes the counts of false positives, false negatives, true positives, and true negatives.

$$\text{Accuracy Score} = \frac{A + D}{A + B + C + D}$$

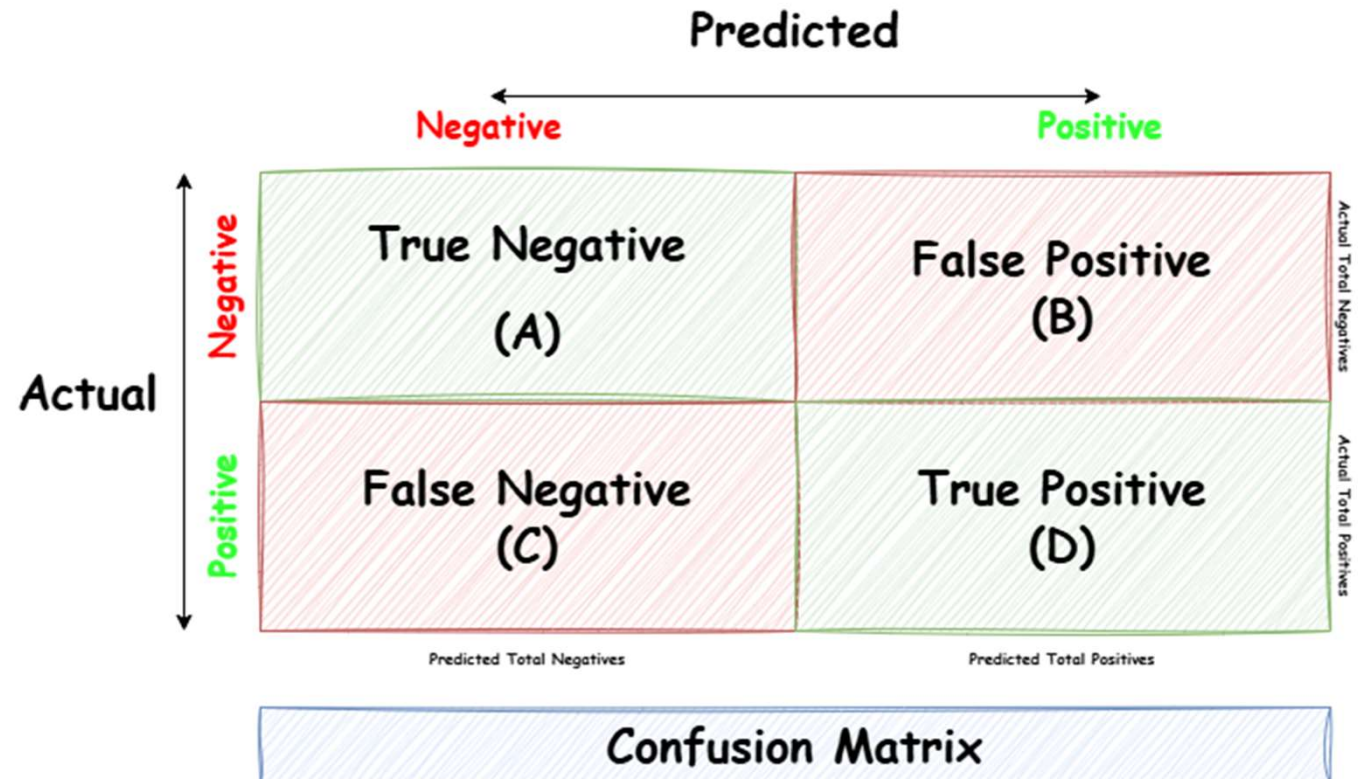


$$A + B + C + D = \text{Total Number of Records}$$

**Will you be happy with High False Positives or High False Negatives
for cancer detection model?**

Confusion Matrix

- **False Positive:** Patient is told they have cancer but they don't. May cause stress, but follow-up tests can confirm they're healthy.
- **False Negative:** Patient has cancer but model predicts they don't. Dangerous, as treatment is delayed.



TPR, FNR, TNR, and FPR

True Positive Rate (TPR): How well the model identifies actual positives.

$$\text{TPR is also known as } Recall = \frac{TP}{TP+FN}$$

False Negative Rate (FNR): What fraction of actual positives the model misses.

$$FNR = \frac{FN}{TP + FN}$$

True Negative Rate (TNR): How well the model identifies actual negatives.

$$Specificity = \frac{TN}{TN + FP}$$

False Positive Rate (FPR): What fraction of negatives the model incorrectly flags as positive.

$$FPR = \frac{FP}{FP + TN}$$

TPR, FNR, TNR, and FPR

Ideal Model

- High TPR and TNR
- Low FNR and FPR

Dumb Model

TPR = 0 (Detects no positives)

TNR = 1 (Detects all negatives)

FNR = 1 (Misses all positives)

FPR = 0 (No false positives)

A model like this would completely fail in detecting fraud or cancer patients.

Precision, Recall and F1-Score

✓ Precision

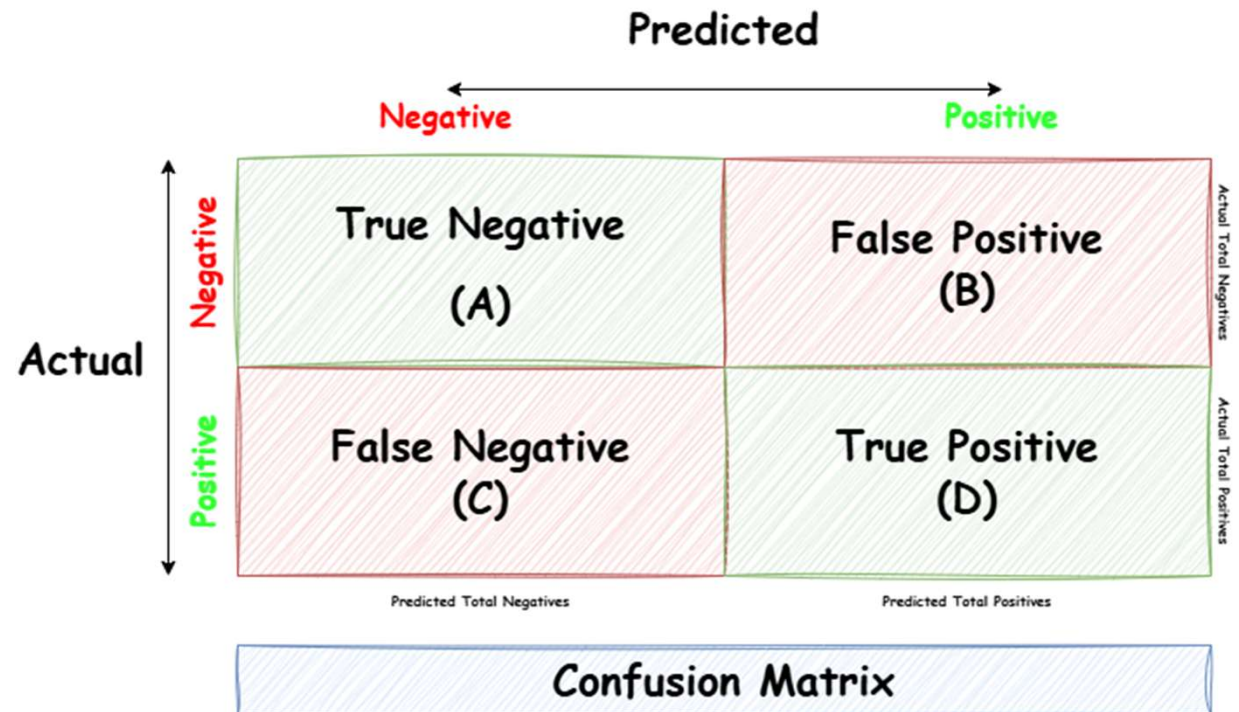
- Definition: How many predicted positives were actually correct?

$$\text{Precision} = \frac{TP}{TP + FP}$$

✓ Recall (Sensitivity)

- Definition: How many actual positives were correctly identified?

$$\text{Recall} = \frac{TP}{TP + FN}$$

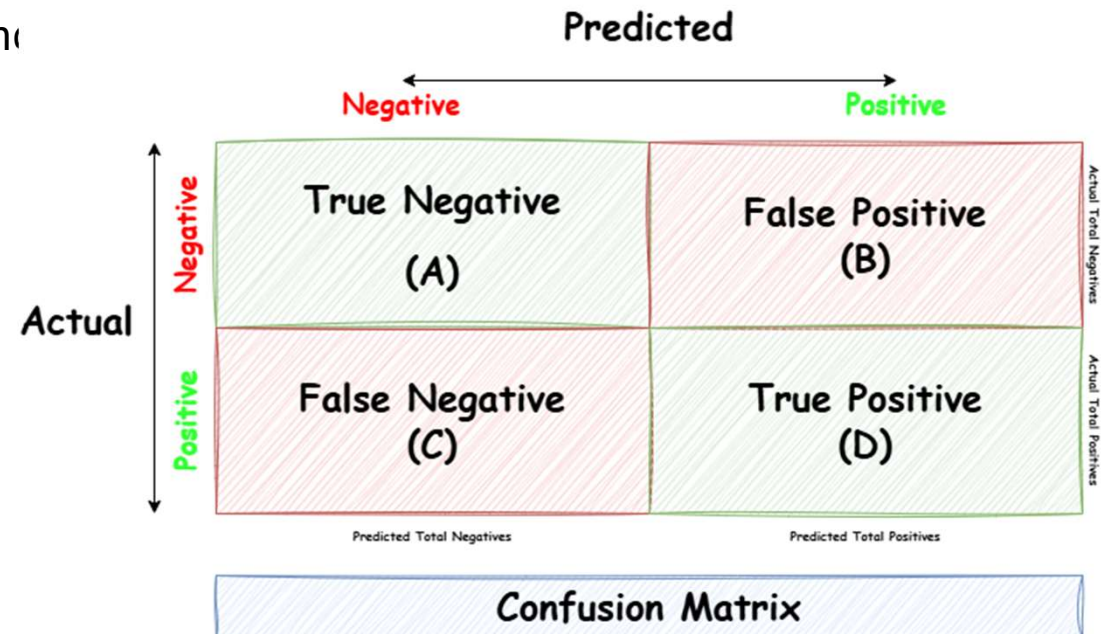


Precision, Recall and F1-Score

✓ F1 Score

Definition: Harmonic mean of precision and recall.

$$F1 - Score = \left(2 * \frac{Precision * Recall}{Precision + Recall} \right)$$



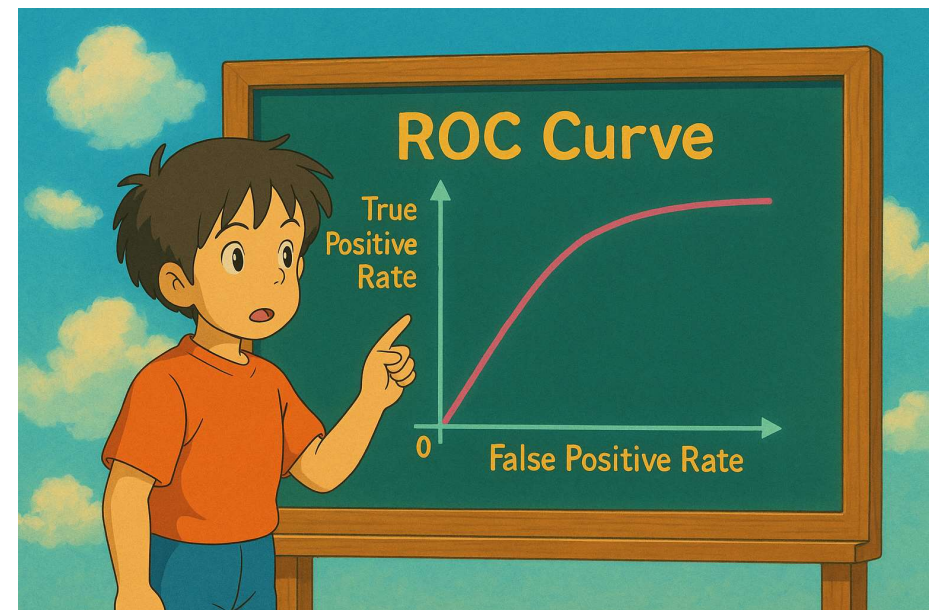
Will you be happy with Low Precision or Low Recall for cancer detection model?

ROC Curve (Area Under the Curve)

An ROC Curve is a plot that shows the performance of a classification model at all classification thresholds. It compares:

- True Positive Rate (TPR) = Recall = $TP / (TP + FN)$
- False Positive Rate (FPR) = $FP / (FP + TN)$

Each point on the ROC curve represents a (FPR, TPR) pair at a specific threshold.



ROC Curve (Area Under the Curve)

 **Let's take a small dataset example**

Suppose we have 5 data points with actual labels (Y) and predicted probabilities (Prob):

So, we have:

Positive class (Actual 1): IDs 1, 2, 5 → Total = 3

Negative class (Actual 0): IDs 3, 4 → Total = 2

ID	Actual (Y)	Predicated Probability
1	1	0.95
2	1	0.85
3	0	0.75
4	0	0.40
5	1	0.30

ROC Curve (Area Under the Curve)



Step 1: Sort by predicted probability (descending)

ID	Actual (Y)	Predicated Probability
1	1	0.95
2	1	0.85
3	0	0.75
4	0	0.40
5	1	0.30

ROC Curve (Area Under the Curve)

Step 2: Choose thresholds

We'll calculate TPR and FPR at each threshold by sweeping through unique predicted probabilities:

Thresholds: ∞ , 0.95, 0.85, 0.75, 0.40, 0.30, 0

We'll count how many are predicted as positive ($\geq$ threshold) and calculate TPR and FPR at each step.

ROC Curve (Area Under the Curve)

Step 3: Calculate TPR & FPR at each threshold

Threshold	Predicted Positives	TP	FP	FN	TN	TPR (Recall)	FPR
∞	0	0	0	3	2	$0/3 = 0$	$0/2 = 0$
0.95	1 (ID 1)	1	0	2	2	$1/3 = 0.33$	$0/2 = 0$
0.85	2 (IDs 1,2)	2	0	1	2	$2/3 = 0.67$	$0/2 = 0$
0.75	3 (IDs 1,2,3)	2	1	1	1	$2/3 = 0.67$	$1/2 = 0.5$
0.40	4 (IDs 1,2,3,4)	2	2	1	0	$2/3 = 0.67$	$2/2 = 1$
0.30	All 5	3	2	0	0	$3/3 = 1$	$2/2 = 1$
0	All 5	3	2	0	0	$3/3 = 1$	$2/2 = 1$

 **AUC (Area Under Curve):** This is the area under the ROC curve. AUC close to 1.0 indicates a strong model; 0.5 means random guessing.

ROC Curve (Area Under the Curve)

✓ Summary

- ROC Curve shows TPR vs FPR at various thresholds.
- Helps compare classification models even with imbalanced data.
- The closer the curve is to top-left, the better the model.
- AUC = 1: perfect, AUC = 0.5: no better than random.

Sensitivity and Specificity

These are evaluation metrics commonly used in **binary classification**, especially in medical diagnostics, fraud detection, and other high-stakes domains.

✓ 1. Sensitivity (True Positive Rate or Recall)

- ◆ Definition: It tells you how well your model is at identifying actual positive cases.

It answers: “Of all the people who actually have the disease (positive class), how many did we correctly identify?”

$$\text{Sensitivity (Recall)} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Sensitivity and Specificity

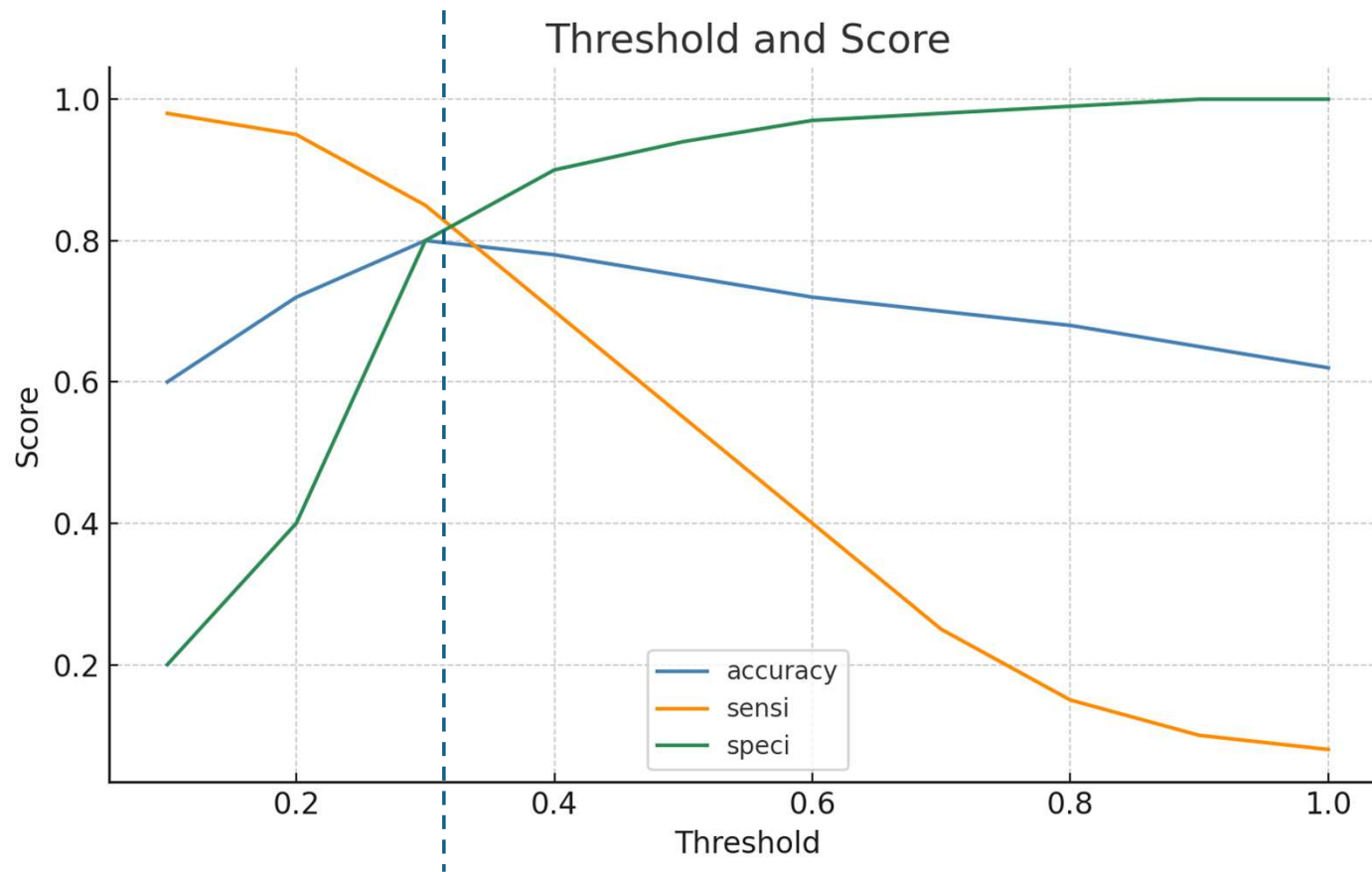
✖ 2. Specificity (True Negative Rate)

- ◆ Definition: It tells you how well your model is at identifying actual negative cases.

It answers: “Of all the people who don’t have the disease, how many did we correctly identify as negative?”

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

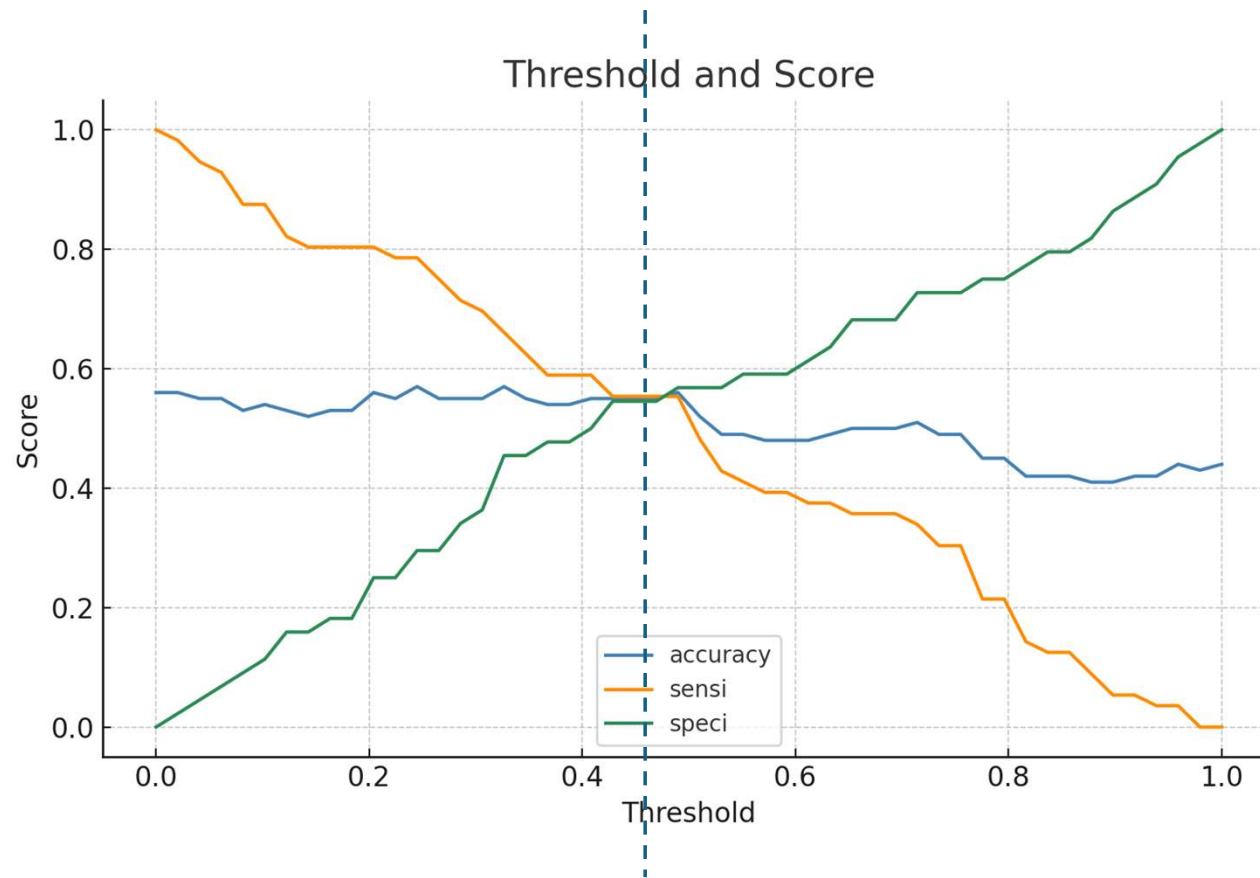
Sensitivity and Specificity



Sensitivity and Specificity

Impact	Explanation
 High sensitivity	Almost all cancer cases are caught (low false negatives).
 Low specificity	More healthy people flagged for cancer (high false positives).
 More people get follow-ups	Extra tests → costlier but safer approach.
 Good for life-critical detection	In cancer, it's better to err on the side of caution.
 Accuracy stays reasonable	Even if specificity drops a bit, overall performance is still solid.

Sensitivity and Specificity

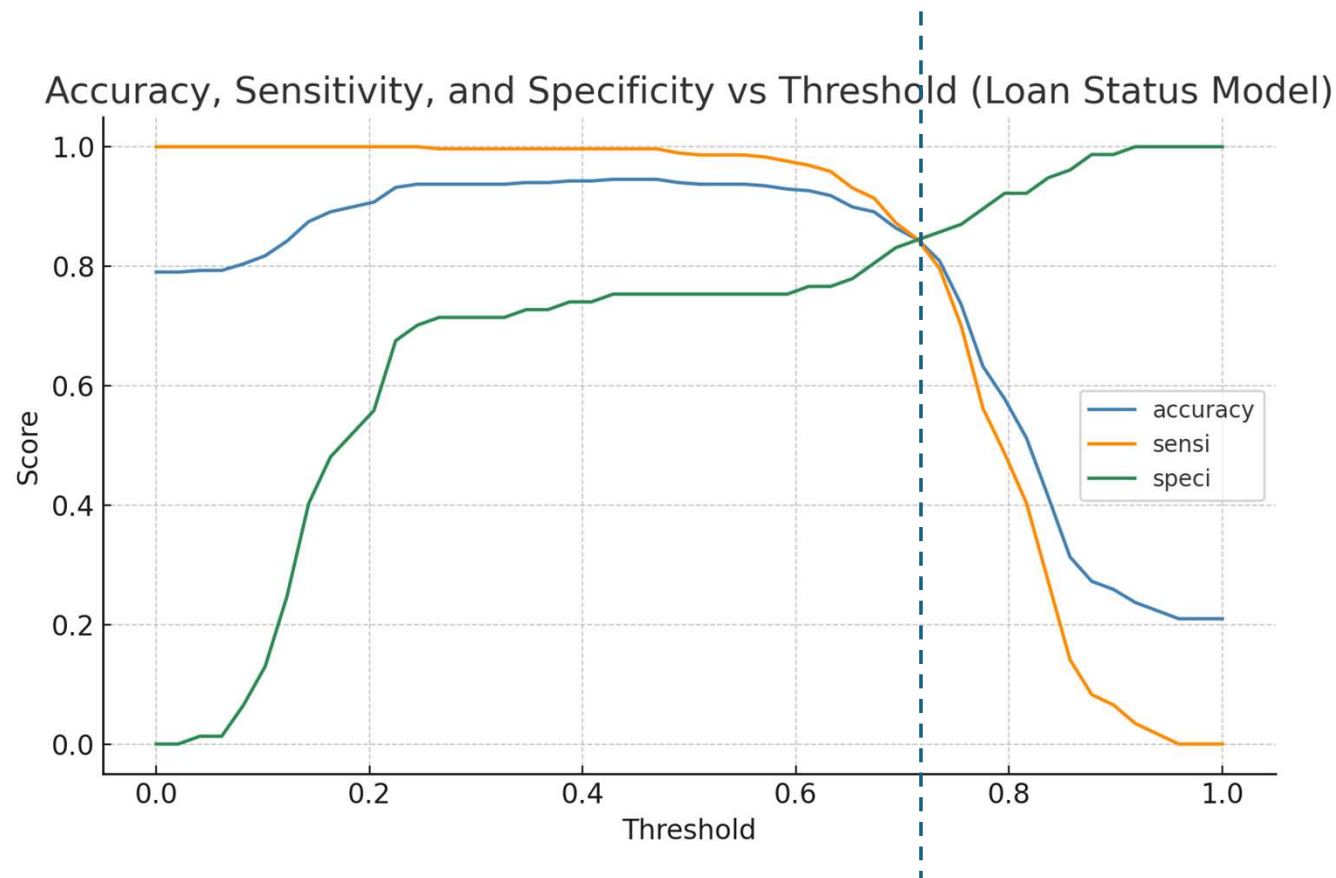


(c) Analytics with Rahul (Rahul Tiwari)

Sensitivity and Specificity

Impact	Explanation
 Sensitivity drops steadily	As threshold increases, the model misses more true positives (e.g., cancer cases).
 Specificity Improves	The model becomes better at ruling out false positives .
 Trade-off becomes clear	Lower sensitivity = higher specificity, and vice versa.
 Accuracy stays fairly stable	The balance of TP and TN keeps accuracy relatively consistent.
 Threshold choice matters	Choose threshold based on whether false positives or false negatives are more costly.

Sensitivity and Specificity



Sensitivity and Specificity

Impact	Explanation
 Sensitivity stays high early	Model confidently catches almost all loan defaulters at low thresholds.
 Specificity rises slowly	It gradually starts identifying good customers correctly as threshold increases.
 Accuracy peaks between 0.4-0.6	There's a sweet spot where model balances between false positives and negatives.
 Beyond threshold 0.6	Model sharply loses sensitivity and accuracy — becomes overly conservative.
 Best threshold 0.4-0.5	Where all three metrics are strong — ideal for balanced business decisions.

Log Loss

What is Log Loss?

While accuracy tells you whether a prediction is right or wrong, log loss tells you how confident your model was in making that prediction.

 Log Loss penalizes false predictions more when the model is overconfident and wrong.

 Log Loss Formula (Binary Classification):

$$\text{Log Loss} = -\frac{1}{N} \sum_1^N [y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)]$$

Where:

y_i = actual label (0 or 1)

p_i = Predicted probability of class 1

N = Number of Observations

Intuition Behind Log Loss

Actual	Predicted Probability	Log Loss (Low = Better)
1	0.99	Very Low (Good)
1	0.60	Low
1	0.10	High (Bad)
0	0.01	Very Low (Good)
0	0.90	High (Bad)

- ✓ If the model assigns high probability to the correct class, log loss is small.
- ✗ If it's confident but wrong, log loss punishes it heavily.

Simple Example for Log Loss

ID	Actual	Predicted Probability	Log Loss Component
1	1	0.9	$-\log(0.9) = 0.105$
2	0	0.8	$-\log(1-0.8) = 1.609$
3	1	0.3	$-\log(0.3) = 1.204$

$$\text{Average Log Loss} = \frac{0.105 + 1.609 + 1.204}{3} = 0.973$$

- ✓ Log Loss is good for probabilistic model.
- ✓ Lower the Log Loss better the model.